

This Month's DVD Contents

In this month's DVD we bring to you interesting, open source electronics design automation, programming and simulation tools, topped with some handy utilities and fun applications. Highlighted below are select tools from the DVD

ABHISHEK A. MUTHA

FidoCadJ (version 0.24.5)

This is a multi-platform vector-drawing program with a complete library of electronic symbols. Schematics and drawings are stored in a very compact text format. There is no netlist concept behind the drawings (so no simulation). Therefore it allows for great graphical flexibility and ease of use, making FidoCadJ the perfect tool for exchanging sketches in forums and news group discussions.

TANGO control system (version 8.1.2)

TANGO is a device-oriented control system for Linux and Windows. It is a toolkit for building object-oriented control systems based on CORBA. It provides a framework in C++, Java and Python for implementing distributed control objects. Python is fully supported for writing TANGO servers and clients at the same level as C++ and Java. TANGO also has a full set of tools and hundreds of device servers.

TinyCAD (version 2.80.08)

TinyCAD is a program for drawing electrical circuit diagrams or schematic drawings. It supports standard and custom symbol libraries and also PCB layout programs with several netlist formats and can produce SPICE simulation netlists. It is also often used to draw one-line diagrams, block diagrams and presentation drawings.

PCB layout tool (version 20140316)

PCB is an interactive PCB editor for Unix, Linux, Windows and Mac systems. It is part of the gEDA suite. It includes a rats nest feature and schematic/netlist import and design rule checking, and can provide industry-

Run all software of DVD with Oracle VirtualBox 5.0

With Oracle VirtualBox 5.0 bundled with this month's DVD, you can now run any software tool and legacy application in a virtual environment. VirtualBox allows you to run more than one operating system (OS) at a time. This way you can run software written for one OS on another (for example, Windows software on Linux or a Mac) without having to reboot to use it. You can run multiple OSes simultaneously like installing an old OS such as DOS or OS/2 even if your real computer's hardware is no longer supported by the OS.

VirtualBox runs on Windows, Linux, Macintosh and Solaris hosts, and supports a large number of guest OSes including, but not limited to, Windows (NT 4.0, 2000, XP, Server 2003, Vista, Windows 7 and Windows 8), DOS/Windows 3.x, Linux (2.4, 2.6 and 3.x), Solaris and OpenSolaris, OS/2 and OpenBSD.

Some popular resources

Apache open office (version 4.1.1). Open source office software suite for wordprocessing, spreadsheets, presentations, graphics and databases

ClamWin (version 0.98.7). Free antivirus used by more than 600,000 users worldwide

Expert PDF Reader. Free PDF viewer software that lets you view and print PDF documents on Windows OSes

VirtualBox (version 5.0). x86 and AMD64/Intel64 virtualisation product for enterprise as well as home use

WinDirStat (version 1.1.2). Disk usage statistics viewer and clean-up tool for Windows

Wireshark (version 1.12.6). Network packet analyser that captures network packets and tries to display that packet data as detailed as possible

standard RS-274X (Gerber), NC drill and centroid data (X-Y data) output for use in the board fabrication and assembly process as well as photo-realistic and design review images. It offers high-end features such as an autorouter and trace optimiser, which can tremendously reduce layout time.

Kactus2 (version 2.8.49)

Kactus2 is the first graphical open source IP-XACT toolset for designing embedded products, especially FPGA based multi-processor systems-on-chips. It provides easier IP reusability and practical hardware/software (HW/SW) abstraction for easier application SW development. You do not lock-in with Kactus2, because you own data and can freely access it at any time. It comes with a core for managing components and designs plus plugins for automation.

gpsim, the gnupic simulator

gpsim is an open source simulator

for Microchip's PIC microcontrollers (MCUs). It supports all three families of PICs: 12-bit, 14-bit and 16-bit cores. You can test the entire PIC for accuracy, from the core to the input/output (I/O) pins and including all internal peripherals. Thus it is possible to create stimuli and tie these to the I/O pins and test the PIC the same way you would in the real world. The standard simulation paradigm including breakpoints, single stepping, disassembling, memory inspect and change and so on has been implemented.

PICsim, the PIC MCU simulator

PICsim emulates MCU PIC16F628/16F877A/18F452 and peripherals such as Universal Synchronous Asynchronous Receiver Transmitter and timers. The simulator architecture permits easy implementation of external elements in C language. PicimLab is a real-time emulator of development boards. ●

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